Mariem Ouertani

Network & Telecommunications Engineering Student | Majorante de Promotion

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EDUCATION

Computer Networks and Telecommunication Engineering, INSAT 09/2025 – Present | Tunis, Tunisia

Integrated Preparatory Cycle, National entrance to engineering schools, INSAT 09/2023 – 05/2025 | Tunis, Tunisia
With Honors,
Ranked 3rd out of 86 (Top of Class)

PROFESSIONAL EXPERIENCE

RPA Developer, SAP Consultant (Internship), Iliade Consulting 07/2025 – 08/2025 | Les Berges du Lac, Tunisie
•Designed and developed an invoice automation robot in SAP IS-U using UiPath.
•Applied artificial intelligence and computer vision techniques to optimize interactions with the SAP interface.
•Project analysis and management.

PROJECTS

Adversarial Neural Cryptography, Personal Research Project (PPP) 01/2026 – Present
- Implemented the **"Alice-Bob-Eve" architecture** based on the research paper "Learning to Protect Communications with Adversarial Neural Cryptography".
- Developed an end-to-end encryption system using **GANs** to learn robust protocols without classical algorithms.
- **Stack:** Python, PyTorch, NumPy, Scikit-learn.

Cybersecurity Audit & Mobile Traffic Analysis (Android) , Academic project 09/2025 – 12/2025
- **Expertise:** Conducted an in-depth study of network traffic monitoring and application metadata exposure.
- **Key Achievement:** Performed traffic auditing and packet capture using **Wireshark/TShark** to identify vulnerabilities in modern protocols (**QUIC, TLS 1.3, MQTT**).
- **Impact:** Defined security countermeasures and **Privacy by Design** recommendations to protect user data.

Smart Agri-Sensor: IoT & Edge AI Solution , Academic project 11/2025 – 12/2025
- **Expertise:** Designed an end-to-end IoT solution for precision irrigation and environmental monitoring.
- **Key Achievement:** Implemented **Edge AI** (Rule-Based Logic) on an ESP32 microcontroller for autonomous crop status classification.
- **Stack:** C++, MQTT Protocol (EMQX), Node-RED Dashboard, and low-power optimization (Deep Sleep).

Supply Chain Optimization Dashboard (Operations Research) , Academic projet 10/2025 – 12/2025
- **Expertise:** Translated complex industrial constraints into profitable strategic decisions.
- **Key Achievement:** Developed mathematical models for production optimization and Max-Flow algorithms using **Advanced Python**.
- **Impact:** Built a decision-making interface to simulate production scenarios and visualize real-time impact on **gross margin**.

REASERCH & IMPLEMENTATION

Next-Gen Networking Simulation (6G & Wi-Fi 7)
Python simulation of **OFDMA**, MU-MIMO, and **Federated Learning** for channel quality prediction in 6G networks.

Deep Learning & Transformers
- Re-implementation of the **"Attention Is All You Need"** paper (Transformers) for automatic modulation classification in 5G.
- Ongoing experimentation with **AI Agents** and attention mechanisms for system optimization.

SKILLS

Languages: Python (Expert), C/C++, Java, JavaScript, PHP. | **Networking:** 5G/6G, Wi-Fi 7, TCP/IP, CCNA, Wireshark. | **AI & Data:** PyTorch, TensorFlow, Scikit-learn, GANs, LLMs. | **Tools:** Git, Docker, Linux (Ubuntu/Debian), SAP, UiPath. | **Full Stack Development** |
Communication, Teamwork, Time Management, Agile Scrum, Organization, Critical Thinking, Problem Solving

CERTIFICATES

Blockchain — Hashgraph Developer (The Hashgraph Association). | **IT & AI — 50+ credentials from NVIDIA, IBM, Google, AWS, and Cisco (CCNA, Cybersecurity, Cloud). Full list and verification links available on my LinkedIn profile.**

LANGUAGES

Arab — Native • **French** — Bilingual • **English** — Fluent , C1 Advanced • **German** — B1

ORGANISATIONS

Junior Entreprise , 09/2024 – Present | Tunis, Tunisia
Member in Project Pole (Web Development Cell) and Commercial Development team
.Actively contributed to the realization of various projects.
.Actively contributed to the organization of major events such as the Junior Enterprise INSAT Forum 2024 & 2025 , JCR 1.0 and 2.0 and Seneca Hack.

IEEE INSAT SB , Active Member 09/2023 – Present | Tunis, Tunisia
Managed SPA'x events and served as Tomorrow Land Ambassador.

IEEE Computer Society Chapter INSAT SB, 01/2025 – 05/2025 | Tunis, Tunisia

Member in Project Development Department

.Actively contributed to the realization of the INSAT Platform project.